

AI Assisted Coding

Assignment – 16.3

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Batch: 45

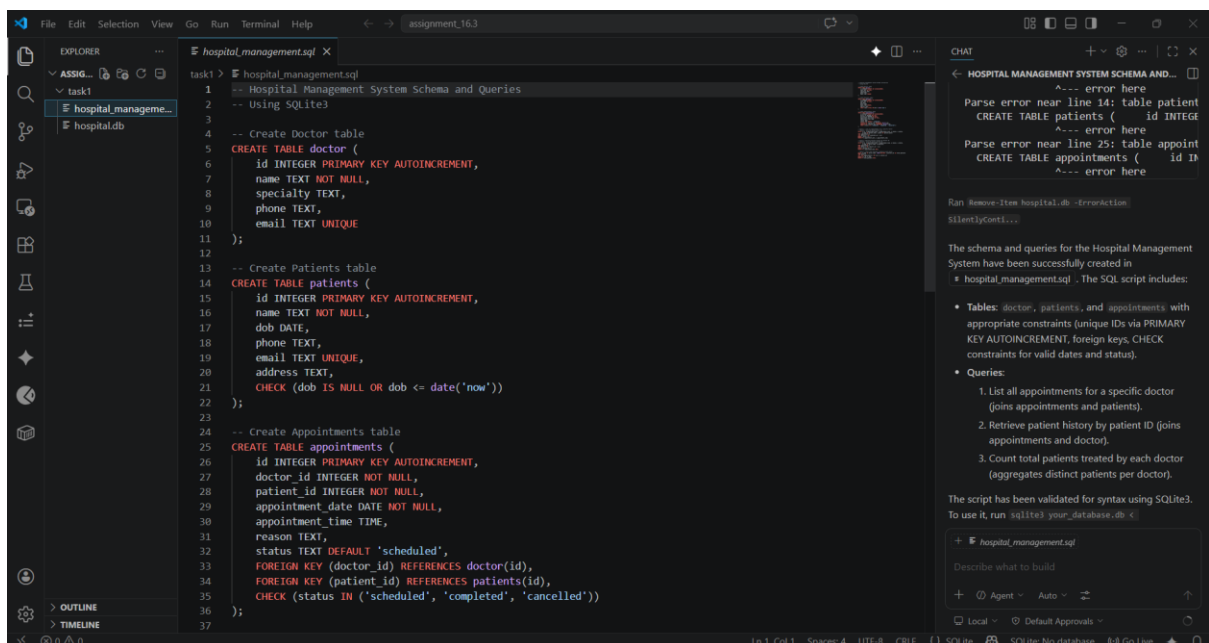
Task1: Create schema and queries for a Hospital Management System.

Instructions:

- Tables: Doctors, Patients, Appointments.
- Use AI to define constraints (unique IDs, valid dates).
- Generate queries:
- List all appointments for a specific doctor.
- Retrieve patient history by patient ID.
- Count total patients treated by each doctor.

Expected Output:

- Normalized schema and SQL queries with joins.



The screenshot displays a code editor with a file explorer on the left showing 'task1' and 'hospital_management.sql'. The main editor contains the following SQL code:

```
task1 > hospital_management.sql
1  Hospital Management System Schema and Queries
2  -- Using SQLite3
3
4  -- Create Doctor table
5  CREATE TABLE doctor (
6      id INTEGER PRIMARY KEY AUTOINCREMENT,
7      name TEXT NOT NULL,
8      specialty TEXT,
9      phone TEXT,
10     email TEXT UNIQUE
11 );
12
13 -- Create Patients table
14 CREATE TABLE patients (
15     id INTEGER PRIMARY KEY AUTOINCREMENT,
16     name TEXT NOT NULL,
17     dob DATE,
18     phone TEXT,
19     email TEXT UNIQUE,
20     address TEXT,
21     CHECK (dob IS NULL OR dob <= date('now'))
22 );
23
24 -- Create Appointments table
25 CREATE TABLE appointments (
26     id INTEGER PRIMARY KEY AUTOINCREMENT,
27     doctor_id INTEGER NOT NULL,
28     patient_id INTEGER NOT NULL,
29     appointment_date DATE NOT NULL,
30     appointment_time TIME,
31     reason TEXT,
32     status TEXT DEFAULT 'scheduled',
33     FOREIGN KEY (doctor_id) REFERENCES doctor(id),
34     FOREIGN KEY (patient_id) REFERENCES patients(id),
35     CHECK (status IN ('scheduled', 'completed', 'cancelled'))
36 );
37
```

The right-hand side of the image shows an AI chat window titled 'HOSPITAL MANAGEMENT SYSTEM SCHEMA AND...'. It contains the following text:

Parse error near line 14: table patient
^--- error here
CREATE TABLE patients (id INTEGE
^--- error here
Parse error near line 25: table appoint
CREATE TABLE appointments (id IN
^--- error here

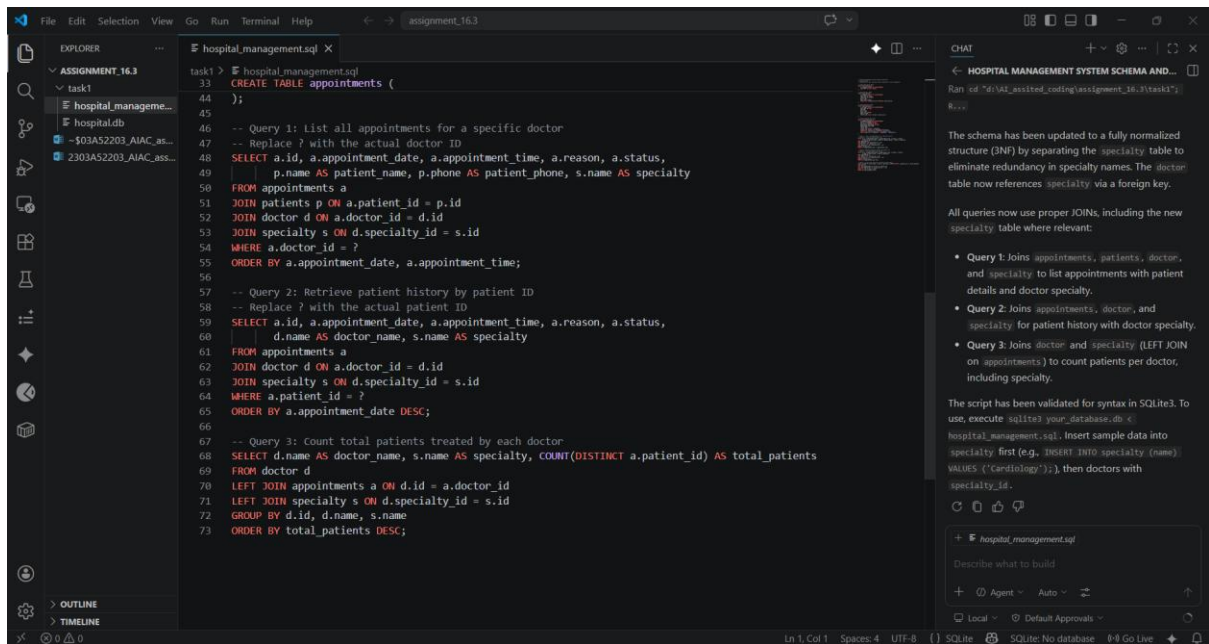
Ran Remove-Item hospital.db -ErrorAction:
SilentlyContinue...

The schema and queries for the Hospital Management System have been successfully created in # hospital_management.sql . The SQL script includes:

- Tables: doctor, patients, and appointments with appropriate constraints (unique IDs via PRIMARY KEY AUTOINCREMENT, foreign keys, CHECK constraints for valid dates and status).
- Queries:
 1. List all appointments for a specific doctor (joins appointments and patients).
 2. Retrieve patient history by patient ID (joins appointments and doctor).
 3. Count total patients treated by each doctor (aggregates distinct patients per doctor).

The script has been validated for syntax using SQLite3. To use it, run `sqlite3 your_database.db <`

Below the chat window, there is a section for 'hospital_management.sql' with a 'Describe what to build' prompt and buttons for 'Agent', 'Auto', and 'Default Approvals'.



Task 2 – Real-Time Application: Online Shopping Database

Scenario:

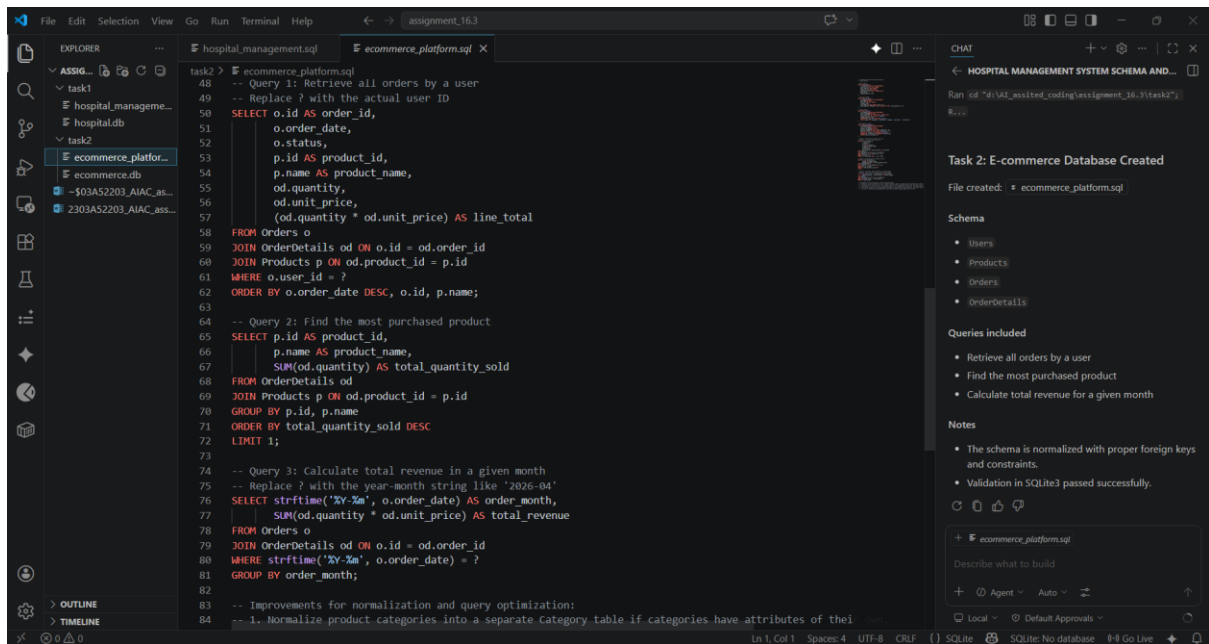
Design a database for an E-commerce platform.

Requirements:

- Tables: Users, Products, Orders, OrderDetails.
- Generate queries to:
- Retrieve all orders by a user.
- Find the most purchased product.
- Calculate total revenue in a given month.
- AI should also suggest normalization improvements and query optimization.

Expected Output:

- Complete schema with relationships + SQL queries for analytics.



Task 3 – Library Database

Task:

Design schema for a Library Management System.

Instructions:

- Tables: Books, Members, Loans.
- Use AI to suggest indexing strategy for faster queries.
- Generate queries:
- Retrieve all books currently issued.
- Find overdue books (loan date > 30 days).
- Count number of books loaned by each member.

Expected Output:

- Schema + SQL queries demonstrating joins and conditions.

task3 > library_system.sql X

EXPLORER

ASSIGNMENT 16.3

task1

hospital_management_system.db

task2

ecommerce_platform.db

task3

library_system.sql

library.db

~\$03A52203_AIAC_as...

2303A52203_AIAC_ass...

OUTLINE

TIMELINE

task3 > library_system.sql

```
47 -- Query 1: Retrieve all books currently issued
48 SELECT l.id AS loan_id,
49        b.id AS book_id,
50        b.title,
51        b.author,
52        m.id AS member_id,
53        m.name AS member_name,
54        l.loan_date
55 FROM Loans l
56 JOIN Books b ON l.book_id = b.id
57 JOIN Members m ON l.member_id = m.id
58 WHERE l.status = 'issued'
59 ORDER BY l.loan_date DESC;
60
61 -- Query 2: Find overdue books (loan date > 30 days and not returned)
62 SELECT l.id AS loan_id,
63        b.id AS book_id,
64        b.title,
65        m.id AS member_id,
66        m.name AS member_name,
67        l.loan_date,
68        julianday('now') - julianday(l.loan_date) AS days_since_loan
69 FROM Loans l
70 JOIN Books b ON l.book_id = b.id
71 JOIN Members m ON l.member_id = m.id
72 WHERE l.status = 'issued'
73       AND julianday('now') - julianday(l.loan_date) > 30
74 ORDER BY days_since_loan DESC;
75
76 -- Query 3: Count number of books loaned by each member
77 SELECT m.id AS member_id,
78        m.name AS member_name,
79        COUNT(l.id) AS total_loans
80 FROM Members m
81 LEFT JOIN Loans l ON m.id = l.member_id
82 GROUP BY m.id, m.name
83 ORDER BY total_loans DESC;
```

CHAT

HOSPITAL MANAGEMENT SYSTEM SCHEMA AND...

Summarizing outputs clearly

Task 3 Completed

File created: library_system.sql

What's included

- Tables: Books, Members, Loans
- Index strategy:
 - idx_loans_book_id on Loans(book_id)
 - idx_loans_member_id on Loans(member_id)
 - idx_loans_status_loan_date on Loans(status, loan_date)
 - idx_books_isbn on Books(isbn)

Queries generated

- Retrieve all books currently issued
- Find overdue books (loan_date > 30 days)
- Count books loaned by each member

Validation

- Ran successfully in SQLite3 with no errors
- Command output: none

library_system.sql

Describe what to build

Local Agent Auto

Ln 1, Col 1 Spaces: 4 UTF-8 CR LF 1 SQLite 65 SQLite No database 84 Go Live